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Biomimetic of a hummingbird: Design and analysis of wing and flapping mechanism

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Introduction:

Our present work focuses on analyzing flight path, required flapping rate, and kinematic study of a mechanism to achieve the flight path that a hummingbird generally exhibits. There is a portion of weight approximation and wing analysis; this is a part of the project works as it provides a variety of input parameters for consideration during the kinematic design of the flapping mechanism.

Objectives:

Following the design procedure of biomimicry the design procedure is split into the following stages:

- Photogrammetry and Kinematic Path of Flight
- Weight Approximation
- Wing Design and Analysis
- Mechanism Design and Analysis

Two strokes plotted side by side:



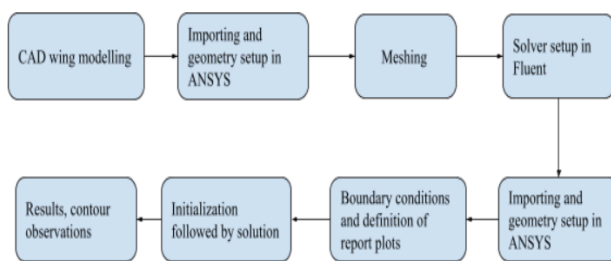
Front view of flapping:



Electronics Weight Approximation for Test Bed:

Sl.No	Name	Type	Voltage/Ampere per Hour	Weight(grams)
1	MPU-9520	Inertial Measurement Unit	3.3/60 mAh	4g
2	Neo-6M	Antenna	-	2g
3	Coreless Motor	Rotary Motor	3.3V	20g

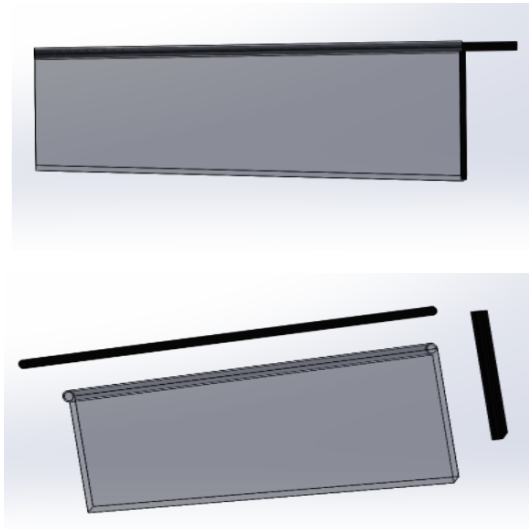
A brief procedure to carry out the steady-state CFD simulation:



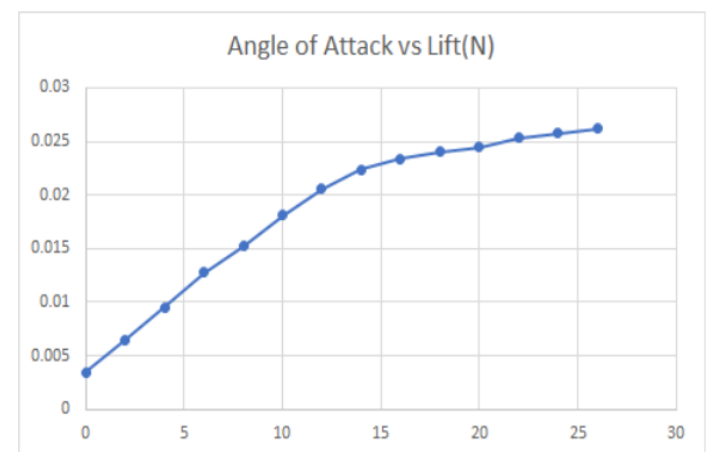
Wing lift formula

- Frequency induced flapping lift
- Flat and rigid wing
- $Lift = \rho C_L S (\Phi f R)^2 \text{ (mN)}$

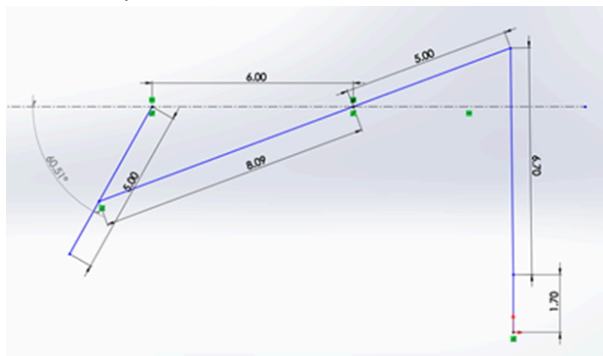
Wing CAD modelling:



Lift vs Angle of attack plot:

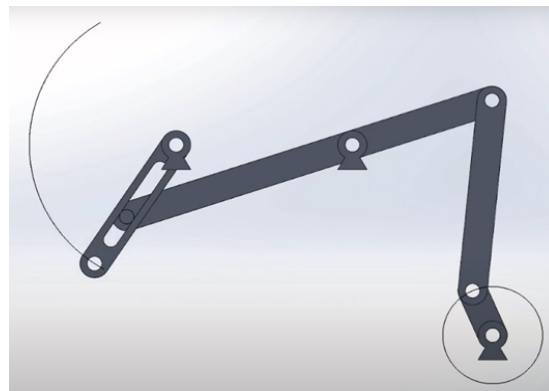


Free body diagram and transmission angles(of 4-bar component):

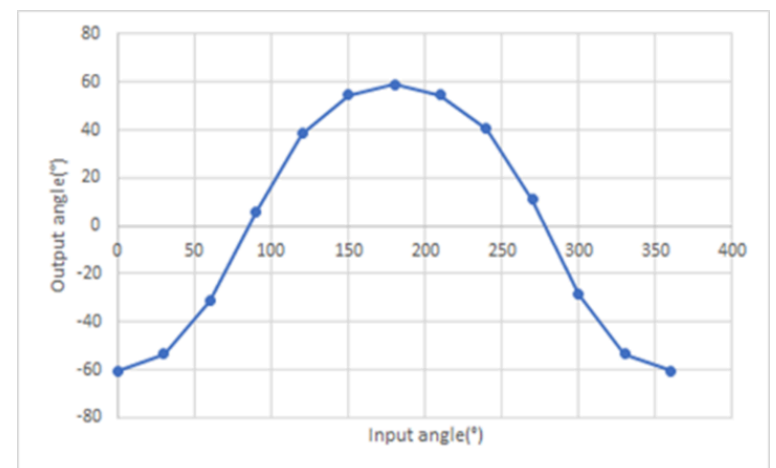


$$\eta_{\max} \approx 114^\circ$$
$$\eta_{\min} \approx 65^\circ$$

Conceptual model:



Position, velocity and acceleration graphs:



Results and conclusions:

- Kinematic path analysis is carried out using image processing
- The total testbed weight is approximated to be 52 grams
- The optimal angle of attack is in the range of 10-15 degrees
- Wing dimensions; Wing length = 82.5mm, tip chord = 22mm, Flapping frequency = 70Hz
- The mechanism gives a flapping amplitude of $\pm 60^\circ$ with the input angle being 360° . The dimensions mentioned can be scaled to appropriately fit with other components.

Future work:

- Use of better image processing techniques and libraries for enhanced flight path analysis
- An improved aerodynamic study by implementing dynamic mesh analysis with an appropriate UDF
- Mechanism improvement which can achieve the figure of eight motion
- Progressing towards synthesizing spatial mechanism to ensure transmission angle lies in the feasible range

